UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIŚ 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

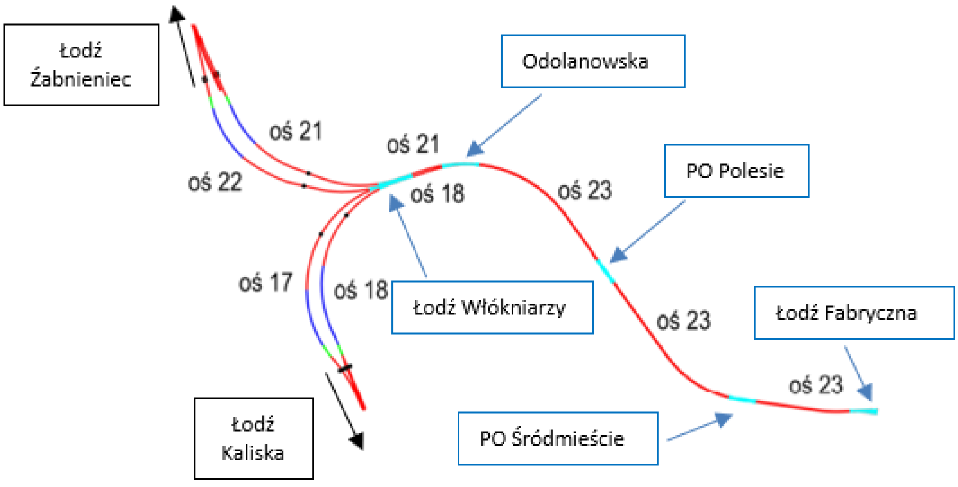

Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 28/08/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+692.09 after the erection of ring n. 229; which corresponds to a total excavation length of 274.38 m from the starting chainage.

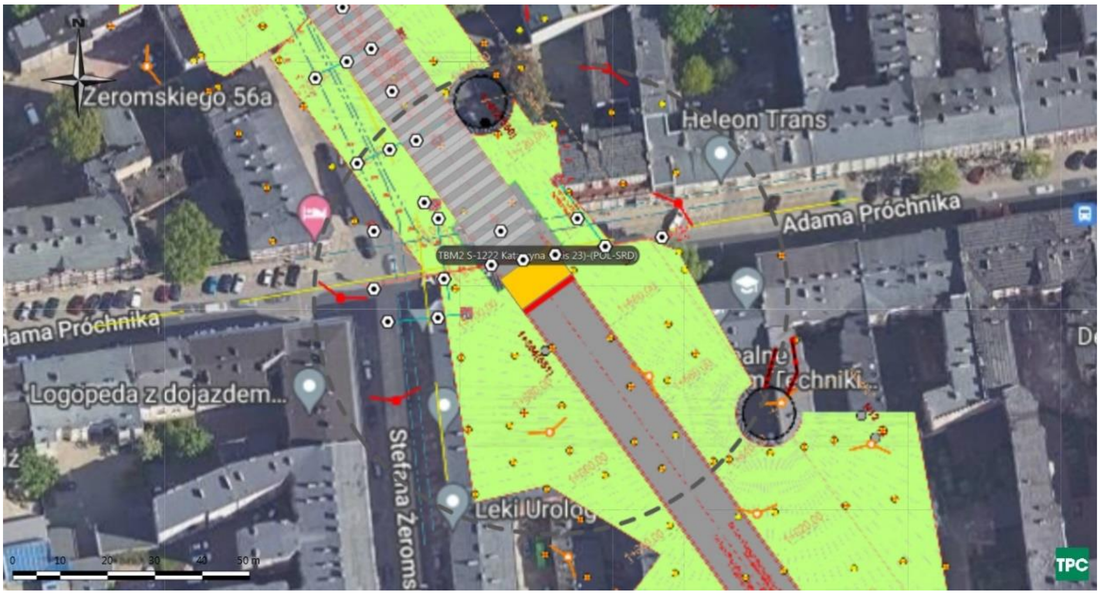

Figure 1-2 Plan view of TBM Position on 29/08/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.7 bar. The value is within the attention values.

Figure 2-1 shows EPB sensors.

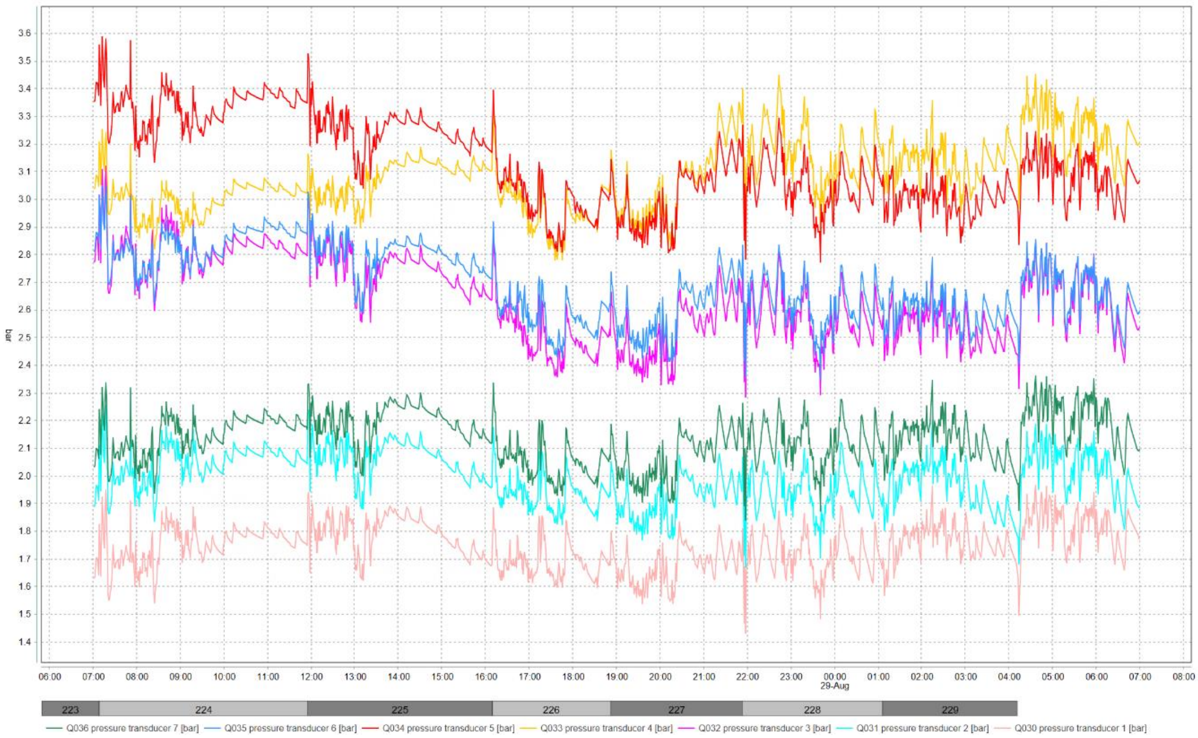

Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.24 bar indicating that the bulk chamber is correctly filled with material. Delta value has a drop at the end of ring 229 and 230, indicating partial void in the upper part of the chamber.

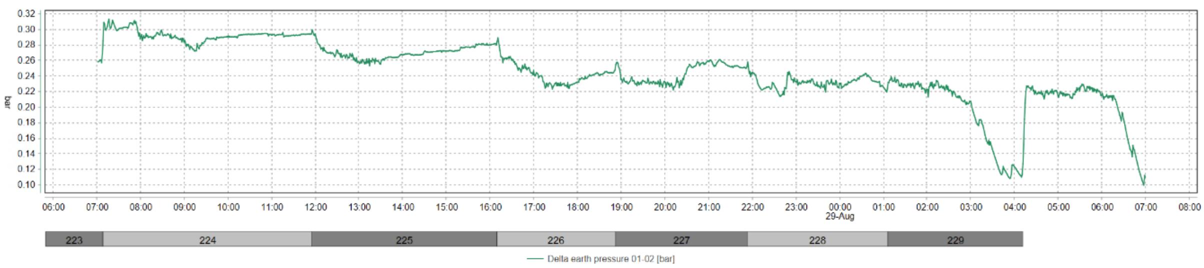

Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 show values much lower than n. 1. The results of Belt scale n. 2 seems to be affected by a constant error and are therefore considered not representative. The belt scale value is a raw indication of the excavated material quantity. The values of excavated material in rings 228 and 229 are higher than expected.

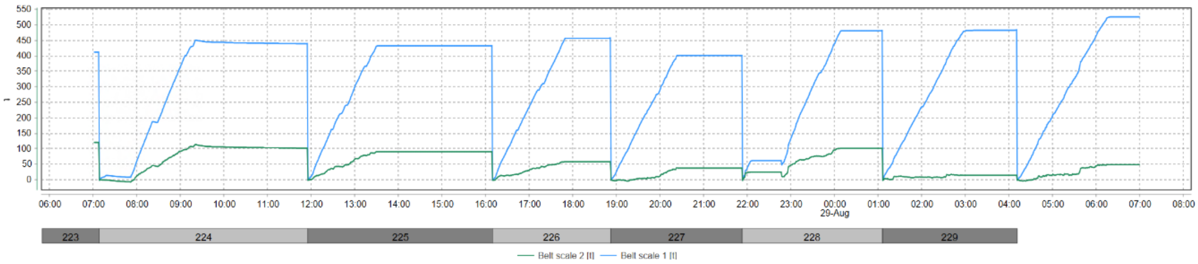

Figure 2-3 Belt scales

Table 2-1 Belt scale values at the end of each advance

Ring	Belt scale 1 (t)	Belt scale 2 (t)
223	414	122
224	452	115
225	434	92
226	459	60
227	403	40
228	483	103
229	486	19

SYSTRA

3. THRUST AND CUTTERHEAD TORQUE

During excavation, the average torque is of 6.5 MNm.

During advancement, average advance rate is of 15 mm/min.

During excavation, thrust force shows an average value of 58.1 MN and a maximum of 81 MN. Thrust value as reduced with reference to last days.

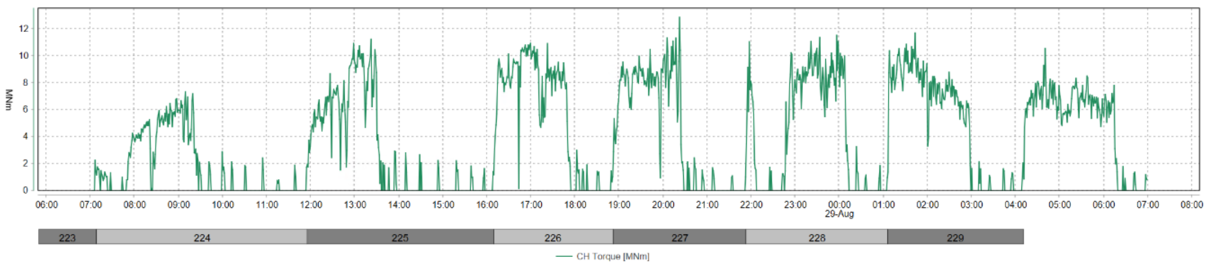

Figure 3-1 CH Torque

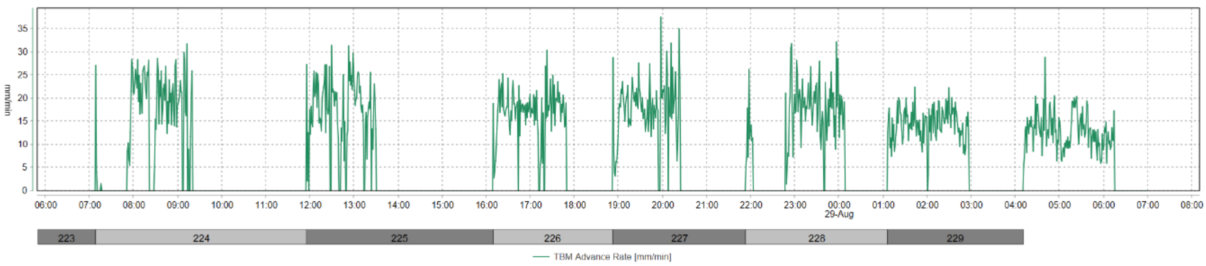

Figure 3-2 Advance rate

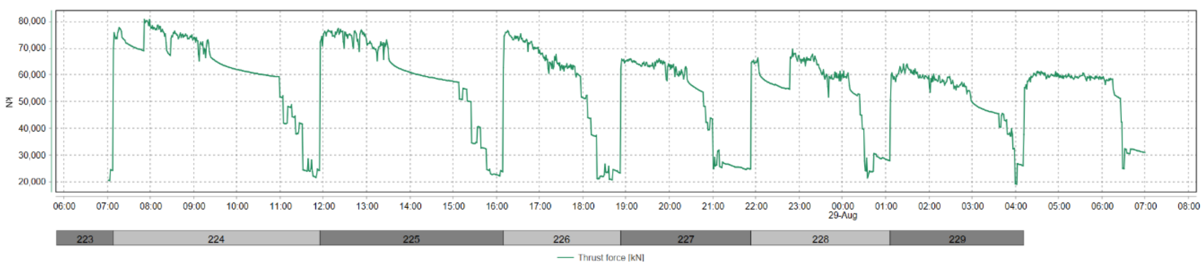

Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection has a volume above the minimum theoretical annular volume +%10 of 13.54 m^3. Ration B/A is at 7%.

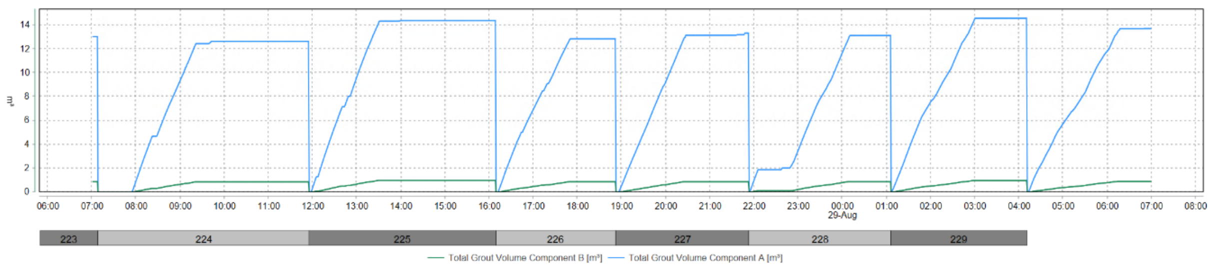

Figure 4-1 Grout volumes

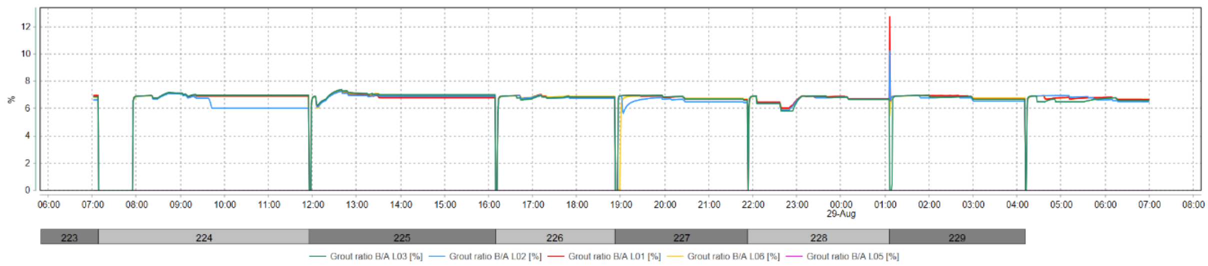

Figure 4-2 Volume ratio Components B/A

Table 4-1 Grout volumes per ring

Ring	Component A (m^3)	Component B (m^3)	B/A (%)
223	13.1	0.9	7
224	12.7	0.9	7
225	14.4	1	7
226	12.9	0.9	7
227	13.4	0.9	7
228	13.2	0.9	7
229	14.6	1	7

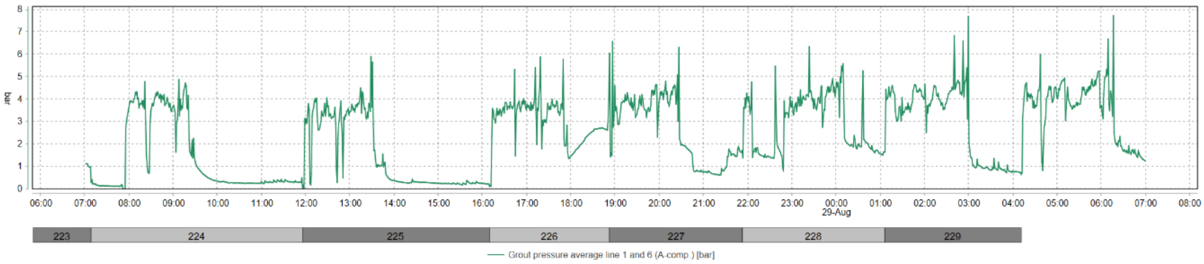

Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about 11.29 mm on the front edge of the shield and the horizontal deviation of about 4.72 mm (Figure 5-1). Maximum radial deviation is of 25.03 mm.

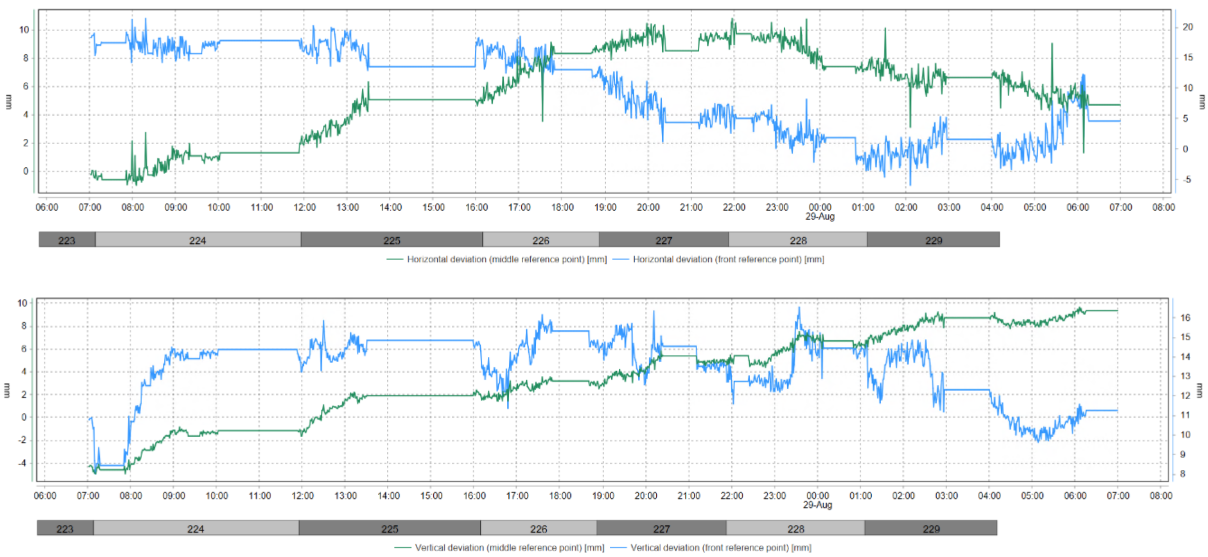

Figure 5-1 Machine deviation from the alignment

SYSTRA

6. MONITORING

According to Geoscope, settlements of the buildings above the tunnel are kept within the limits.

Figure 6-1 Monitoring points

The monitoring points on Zeromskiego street are stable.

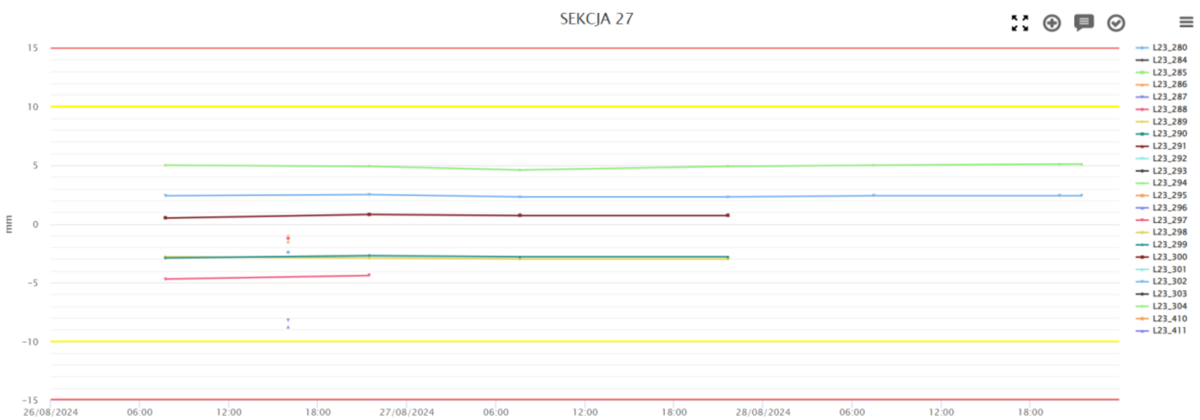

Figure 6-2 Settlement of Section on Zeromskiego street

The monitoring points on Prochnika street show a settlement values of about 3 mm in 48 hours, particularly in point L23_308, L23_309, L23_313 and L23_314, that are the point closest to tunnel axis.

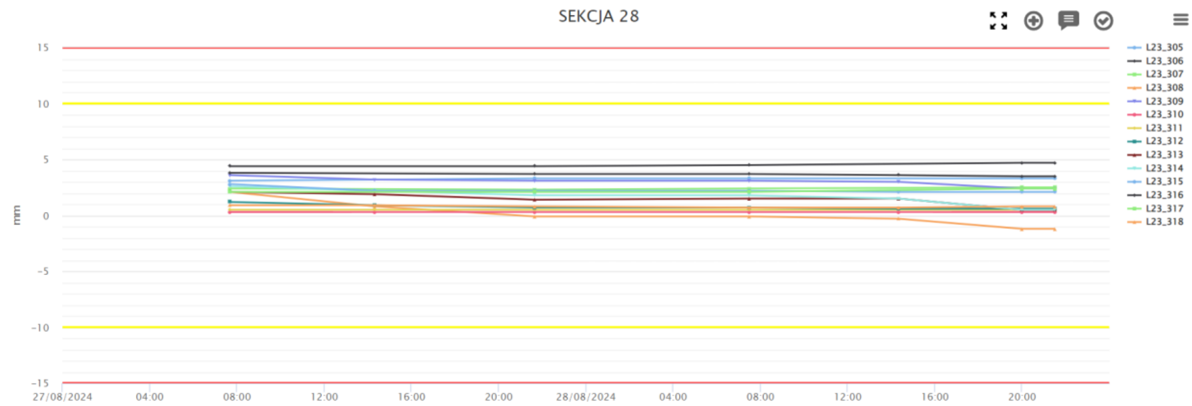

Figure 6-3 Settlement of Section on Prochnika street

On PR0450, monitoring points are stable in the last 24 hours.

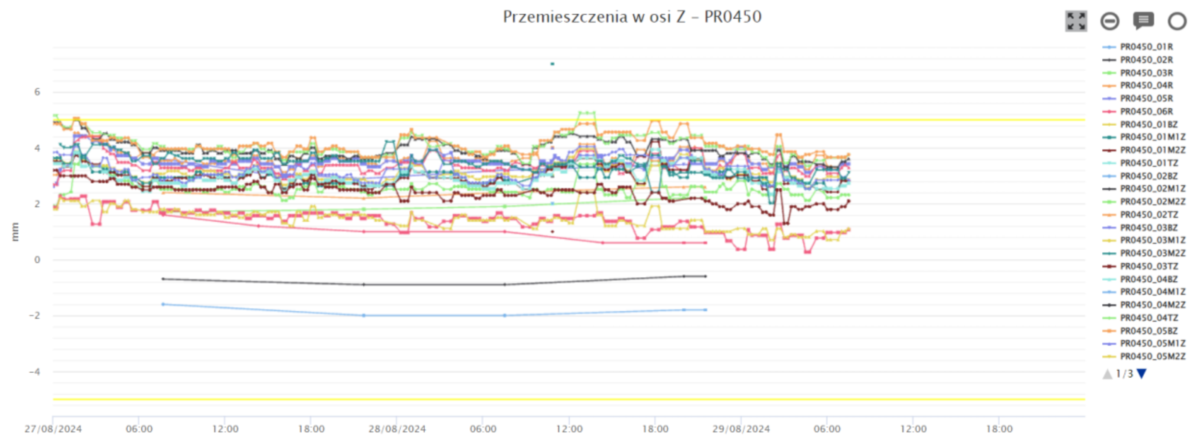

Figure 6-4 Settlement of building PR0450

On PR0430, monitoring points are stable.

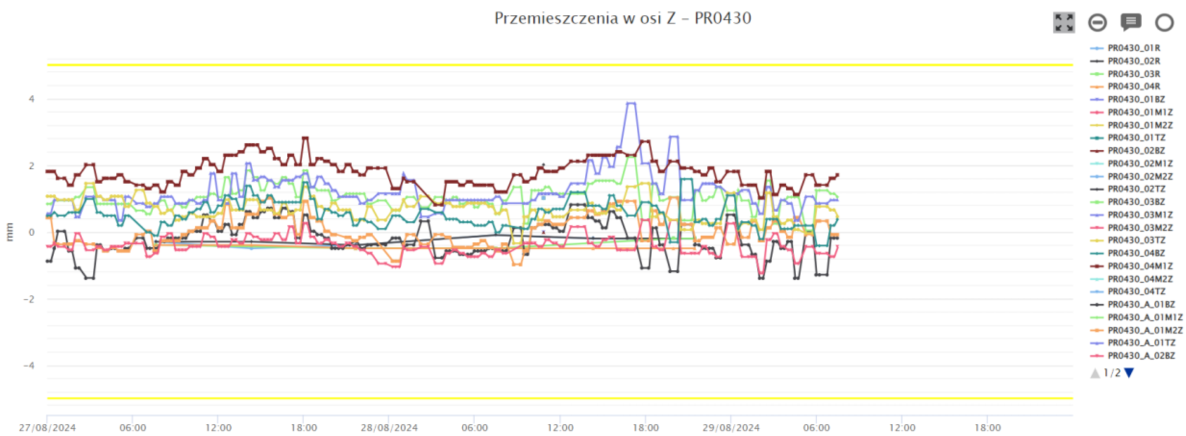

Figure 6-5 Settlement of building PR0430

On PR0460, monitoring points columns 04, 05 and 13, on the road wall, show a downward trend of about 1 mm.

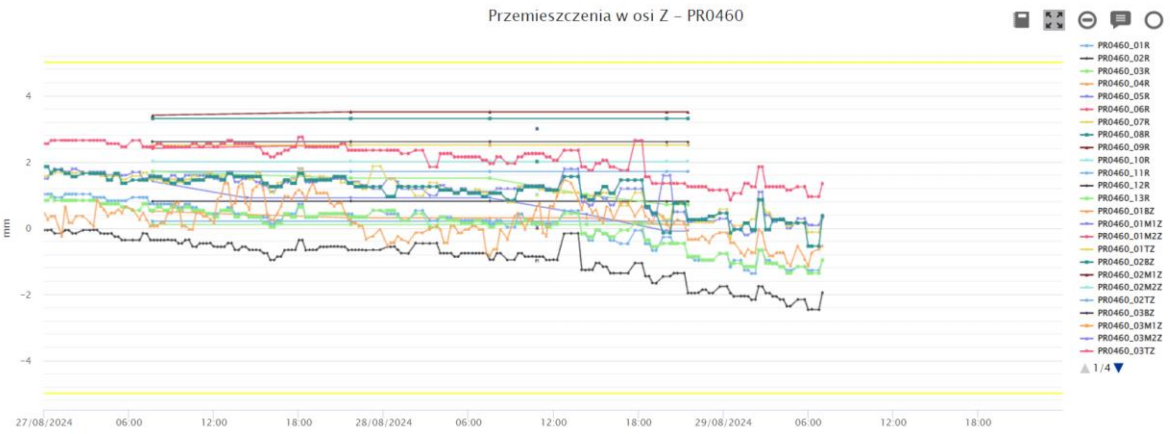

Figure 6-6 Settlement of building PR0460

On PR0440, monitoring points are stable.

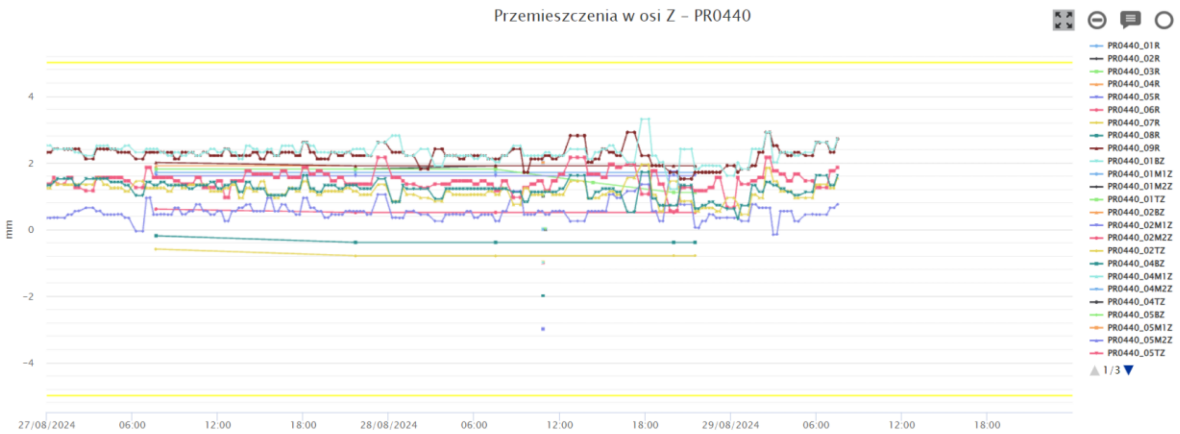

Figure 6-7 Settlement of building PR0440

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 28/08/2024 of Soletanche, compensation grouting has been performed on the 28/08/2024 from Shaft 2 in Próchnika 45.

The total amount of declared injected volume is 2268 l.

According to Raport Dobowy 28/08/2024 of Soletanche, compensation grouting has been performed on the 28/08/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 5801 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure. The FIR value is computed based on the excavated material amount per ring and reported in the Table.

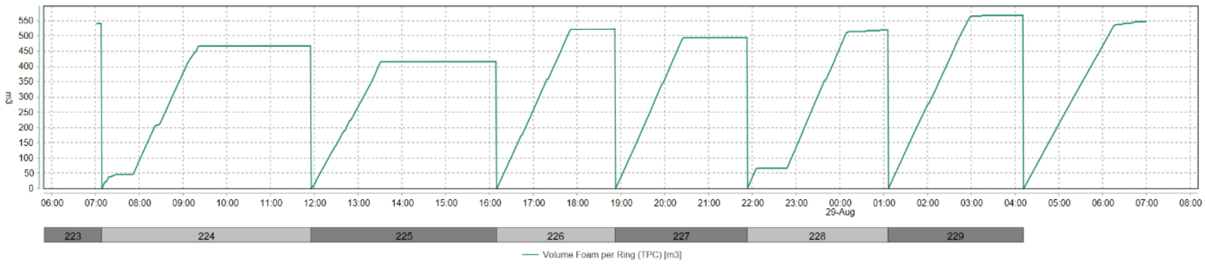

Figure 8-1 Injected foam volume

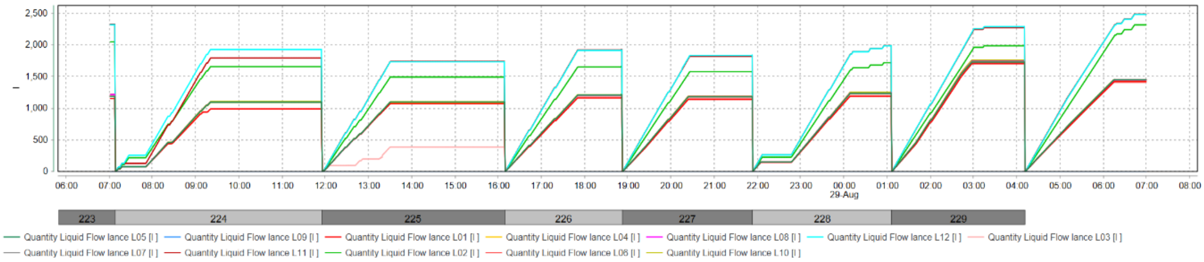

Figure 8-2 Conditioning liquid volume

Table 8-1 Soil conditioning data

Ring	Liquid volume (m^3)	C (%)	FIR (%)
223	16.4	2	255
224	14	2	221
225	13.1	2	196
226	15.2	2	246
227	14.7	2	233
228	15.7	2	245
229	20.5	2	268